David Rock

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PROFESSIONAL SUMMARY

Electrical engineering student graduating in 2027 with coursework in control systems and ongoing study of nonlinear dynamics. Apply control theory through Python-based robotic motion, PID tuning, modeling, and experimental analysis. Interested in developing and testing guidance, navigation, and control systems on real hardware.

EDUCATION

California Polytechnic State University, San Luis Obispo 2023–2027

B.S. in Electrical Engineering

Relevant coursework: Control Systems; PHYS 4428 Nonlinear Dynamics and Chaos (in progress); Vision-Based Robotic Manipulation

CONTROLS AND TECHNICAL SKILLS

Controls and dynamics: PID controller design and tuning, plant/controller block diagrams, transient-response analysis (overshoot, settling time, rise time), forward/inverse kinematics, Jacobians, coordinate transforms, trajectory planning

Programming and simulation: Python, MATLAB, Simulink, C, SystemVerilog, Vivado

Hardware and sensing: OpenManipulator-X, DYNAMIXEL actuators, Intel RealSense D435 RGB-D camera, AprilTags, oscilloscopes, multimeters

SELECTED PROJECTS

Vision-Based Robotic Manipulation | OpenManipulator-X, Python

- Programmed a 4-DOF arm for pick-and-place; planned task-space poses, checked inverse-kinematics reachability, and tested motion on hardware.
- Implemented cubic joint-space and quintic task-space trajectories; compared path accuracy, joint speed, and motion smoothness using measured data.
- Built a Jacobian pseudoinverse velocity controller for waypoint tracking; measured a 59 Hz control loop and 48.2 mm/s realized speed against a 50 mm/s command.
- Calibrated an Intel RealSense D435 camera to the robot frame with AprilTags and coordinate transforms; achieved 2.0 mm mean registration error.

Nonlinear Dynamics Lab | Python, Photogate Data

- Analyzed dripping-faucet time series with return maps and interval histograms to distinguish periodic, alternating, and irregular behavior; documented measurement limits.

PLC Cooling Control | Schneider Electric Modicon PLC

- Designed a three-fan temperature-control system with automated sensing and hysteresis logic in Ladder Diagram and Function Block Diagram.

EXPERIENCE

Hardware Engineering Intern, Nextstera Tech Spring 2024

- Developed and tested sensor hardware for lithium-ion detection devices.

Real Tech Controls Summer 2026

- Fabricated sensor test panels by drilling PVC backplates and mounting industrial sensors and transmitters.

ADDITIONAL

Laboratory: 700+ hours using electronic test equipment; automated performance-data collection with Python and Simulink and prepared technical reports.

Leadership: President, Chabad on Campus SLO (2025–present); coordinated campus initiatives and board logistics.